A-A-0051607(DM)
12 August, 1986

INTERIM COMMERCIAL ITEM DESCRIPTION

RESUSCITATOR-INHALATOR-VENTILATOR

The General Services Administration has authorized the use of this interim commercial item description which was developed by the Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, based upon currently available technical information.

This commercial item description covers a dual mode breathing valve, one for ventilation of apneic patients with 100 percent oxygen and the other to ventilate breathing patients with selectable flow rates and oxygen concentrations. An aspirator shall be included.

Salient characteristics:

The breathing valve system shall include a positive pressure, oxygen powered Resuscitator-Inhalator-Ventilator (R-I-V) which shall provide two separate and independent modes, (1) resuscitator and (2) inhalator, neither of which shall affect the function of the other. The first mode shall provide a system for a manually triggered, selectable (adult/child), positive pressure ventilator for apneic patients by providing 100 percent oxygen concentration via a manually controlled plunger. The second mode shall provide a system for supplying oxygen and entrained ambient air to a breathing patient at manually pre-set variable flow rates, requiring no inspiratory effort by the patient to activate the system. The oxygen concentration shall increase as the flow rate is adjusted higher.

The R-I-V shall include a DISS threaded connector for supply tubing, a 22 mm O.D. by 15 mm I.D. outlet for standard masks and endotracheal tube, and a manual control plunger. The R-I-V shall also include at least 6 feet of supply hose, a regulator preset to 50 psi (± 5 psi) with a double cylinder yoke, one adult mask, one child mask, one aspirator with a plastic collection jar (with tubing and Yankauer tip), handwheels (or cylinder post valve wrench), and a rigid (or semirigid) carrying case suitable to accomodate two noncomponent oxygen cylinders (size D). The R-I-V shall also include a medium concentration transportation, soft, see-thru, mask with tubing (at least 7 feet long). Shall include an additional inhalator valve (oxygen control valve) for independent use with a second patient.

AMSC N/A FSC 6515

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Except as otherwise specified, the R-I-V shall meet the Standards, as appropriate, set by the committee on Cardiopulmonary Resuscitation, National Academy of Science, National Research Council. The resuscitator-inhalator-ventilator shall operate without degradation of performance from source pressures of 40 to 60 psi. It shall include appropriate reducing valve and connectors to operate from both oxygen wall outlet and oxygen cylinder(s). The double yoke shall allow an empty cylinder to be exchanged for a full cylinder while the other cylinder is in use. The breathing valve size shall be such that it may be contained within an envelope having overall basic dimensions of 7.5 cm by 4.5 cm by 6.5 cm. Weight shall not exceed 7.5 oz. The center line of the manually controlled plunger (for resuscitator mode) shall be located approximately 4.5 cm from the forward face of the valve's patient outlet.

For the purpose of being fire resistant, being safe for human ventilation (nonpoisonous), and not being subject to degradation, the materials used that are intended to be in the path of pressurized oxygen shall conform with the appropriate NFPA or ASTM Publication(s). Latest publication for each category shall prevail. Materials that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage (see paragraph "Ambient environmental conditions") shall be protected against such corrosion in a manner that will not prevent compliance with the performance requirements specified. Protective coating(s) that will crack, chip, flake, or scale with age or extremes of specified atmospheric conditions shall not be used. The breathing valve shall be fabricated of all metal excluding seal rings and shall be free of diaphrams. The housings shall resist corrosion by fluids with which they are likely to come in contact.

The R-I-V (including components) shall be designed to meet the performance requirements specified, during subjection (and after being subjected) to extremes of environmental conditions (temperature, humidity, altitude, vibration and mechanical shock).

Standard ambient conditions:

a.	Temperature	$23^{o}C \pm 10^{o}C$ ($73^{o} \pm 18^{o}F$)
b.	Relative humidity	50 percent $\pm$ 30 percent
c.	Atmospheric pressure	750 $\pm$ 50 mm Hg (29.5 $\pm$ 2 in. Hg)

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Ambient environmental conditions: The equipment shall operate properly throughout the following ranges (after subjugation to storage temperature).

a. Temperature:

1. storage: -34°C to 65°C
2. operating: -18°C to 49°C

b. Relative Humidity:

1. 5 percent at 40°C ± 2°C (104°F ± 3.6°F)
2. 95 percent at 40°C ± 2°C (104°F ± 3.6°F)
3. 95 percent at 29.5°C ± 2°C (85°F ± 3.6°F)

c. Altitude: Sea level to 35,000 ft.

Performance characteristics:

1. Resuscitation of adult: Deliver from 100 to 130 LPM of 100 percent oxygen instantaneously with maximum positive inspiratory pressure safety relief of 60 mm Hg (± 5 mm Hg).

2. Resuscitation of child: Deliver reduced flow rate (from adult flow rate) of 100 percent oxygen instantaneously with a maximum positive pressure safety relief of 30 mm Hg ± 5 mm Hg.

3. Inhalation: Increasing the variable airflow control shall increase the oxygen concentration up to 85 percent (± 10 percent) at flow rates from 0 to 15 LPM. Maximum flow rate shall be restricted to not exceed 20 LPM.

4. Aspirator: The aspirator shall be venture type oxygen powered. When supplied with source power of 50 psi, it shall be capable of generating a maximum negative pressure within a range from 270 mm Hg to 430 mm hg. The negative pressure shall be either: a. user controlled from 0 to maximum negative pressure; or b. capable of being user set for continuous maximum negative pressure, and momentary operation (such as plunger operated). The average gas consumption shall not exceed 20 LPM. The collection jar (with its lid) shall be capable of withstanding a vacuum of at least 1,000 mm Hg negative pressure without collapsing. The collection jar shall have a minimum capacity of 16 ounces. The suction hose shall be 1/4 inch ID, clear tubing with a Yankauer suction tip (thumb-operated).

5. Breathing valve self-cleaning feature: Vomitus shall be cleared from the breathing valve (after removing the mask from the patient) by activating the plunger.

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6. Breathing valve sterilizability: The breathing valve shall be capable of being cleaned and sterilized without disassembly via: steam, gas, and cold chemical agents.

7. Breathing valve pressure test: There shall be no damage caused to the breathing valve when subjected to pressure up to 200 psi regardless of setting (child/adult) or plunger activation.

8. Supply hose integrity: The supply hose shall resist damage by abrasion, twisting or pulling. The hose shall be flexible and kink proof. The cold bend or brittle temperature of the material shall not be less than (-) 50°F. The hose shall be at least 6 feet (1.8 m) long, fitted with a female DISS coupling in accordance with CGA-V5. It shall have nominal dimensions of 3/16 inch I.D. by 13/32 inch O.D. The hose shall have a working pressure rating of at least 200 psi and a burst pressure of at least 500 psi.

9. Regulator: The regulator shall be single stage, double yoke type preset to deliver 50 psig ± 5 psi with CGA safety standard double pin-indexed yoke (with check valve) for "D" and "E" size medical oxygen cylinders for double cylinder application. The regulator shall have a 1-1/2 to 2 inch diameter stainless steel Bourdon pressure and content gauge having a 4000 psig test. It shall be calibrated from 0 to 2000 psig in 100 psig increments. Major divisions shall be marked at 500 psig content from zero to full in increments of one-quarter.

10. Face mask (one child and one adult): The face masks shall consist of a transparent body section, a flexible (at full temperature operating range) face seal, and a connector capable of receiving (securely) the patient connector of the breathing valve (ANSI Z79.6).

11. Carrying Case: The carrying case shall be a double cylinder unit fabricated of rigid or semirigid material. It shall provide suitable space to house a R-I-V, aspirator, all components, and two "D" size oxygen cylinders. The total weight of the case with all components therein (less noncomponent oxygen cylinders) shall not exceed 14 pounds. The case shall provide space for assorted emergency instruments and the cube shall not exceed 2.16 cubic feet.

The case shall have two (2) corrosion-resistant carrying handles suitably positioned to allow carrying the case in either the horizontal or vertical position and shall be firmly and securely attached to the case. The case design shall be such that it either; a. allows for removal of the cylinder assembly (two cylinders with yoke, regulator, etc. ready for use) from the case; or b. permits easy removal of the lid to facilitate total operation of the unit while it is attached inside the case. Either design shall provide the operator with easy access (visual and physical) to all controls and gages.

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Closure shall be accomplished by at least two (2) corrosion-resistant suitcase type latching mechanisms. All mounting hardware shall be corrosion-resistant, or suitably protected against corrosion. There shall be a suitable rubber plastic seal (or interlocking lips) between the case lid and case bottom. The inside configuration of the case shall be such that the cylinders are securely held. A molded well or insert in the lid shall securely hold the aspirator and provide space for other medical devices. The case shall be designed to suitably hold and protect the contents. The case and contents shall not sustain any damage, disassembly, or degradation in performance when subjected to the vibration and drop tests specified in Performance Characteristics No. 14 and 15 respectively.

12. Sterilizing. Metal assemblies in the patient gas circuit shall be capable of receiving repeated applications of sterilization via: steam, ethylene oxide gas, and cold chemical disinfectant(s). Other components in the patient gas circuit shall be capable of receiving repeated applications of sterilization via ethylene oxide gas and cold chemical disinfections.

13. Operational stability. The resuscitator shall have a minimum operating life of 100,000 cycles with no degradation in performance and without the necessity for readjustments of any controls which are inaccessible to the operator during normal use. Each cycle shall consist of 0.75 seconds on and 2.25 seconds off when driven by a 50 psi compressed air source.

This test shall be performed in the event that an identical unit has not been successfully tested and accepted by the Government. Regardless of previous testing, the Government reserves the right to require this test to be performed on a unit randomly selected from the quantity of this solicitation/contract. The unit selected and tested in accordance with this requirement shall be painted red and labeled "Not for use. This unit has been tested through 100,000 cycles." It shall be separately delivered to the Government as directed by the contracting office.

14. Vibration. The resuscitator in its case shall withstand, without damage, vibration over a swept frequency of 2-35 Hz with a peak amplitude of 6.4 mm (0.25 in) and a sustained 30 Hz vibration for five (5) minutes in each of three (3) mutually perpendicular planes. Acceleration shall range from 0.9 G to 2.1 G. After vibration, the unit shall meet the performance requirements specified herein.

15. Drop test. The resuscitator shall be capable of withstanding a free fall onto a concrete surface or steel plate from a height of one (1) meter without affecting the performance of the device, or cracking the case.

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16. Altitude: There shall be no degradation in performance when the equipment is operating at an atmospheric pressure of 564.4 mm Hg (22.22 in Hg or 8,000 ft above sea level).

17. Decompression: There shall be no degradation in performance when the equipment is operating under the following conditions:

a. Standard ambient conditions.
b. Atmospheric pressure change to 564.4 mm Hg (22.22 in Hg or 8,000 ft above seal level) at a rate of 500 $\pm$ 200 ft/min.
c. Atmospheric pressure change to 178.7 mm Hg (7.035 in Hg or 35,000 ft above seal level) within sixty (60) seconds.
d. Atmospheric pressure increased to standard ambient conditions at the rate of 5000 $\pm$ 500 ft/min.

18. Disassembly and reassembly: The parts that are to be disassembled/assembled for cleaning/sterilizing shall be shape/size designed for easy identification with illustration(s) of the instructions.

19. Controls: There shall be a minimum number of controls and adjustments which shall be easy to operate and conveniently located.

20. Display: Controls shall be marked, indexed and located in such a manner that the control position can be readily identified.

21. Stability: Controls shall be designed so that the setting, position, or adjustment shall not change when the equipment is subjected to the intended service conditions.

22. Identification: All flow controls and adjustments shall be clearly identified. Labeling shall be in the most simple and direct manner possible and easily readable. The actuating plunger shall be color coded red.

23. Instructions: The unit shall include instructions which shall describe the theory of operation and include operating and sterilizing instructions. They shall also include a list of replacement parts by manufacturer's catalog number, descriptive sketches as appropriate, a statement of warranty, and manufacturer's/distributor's name, address, and phone number.

Workmanship. The breathing valve shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One Resuscitator-Inhalator-Ventilator, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contact or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics.

Examination. Examination shall be conducted to determine compliance with specifcation requirements.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. The unit of product for inspection purposes shall be one Resuscitator-Inhalator-Ventilator. Inspection level shall be S-2 for both examination and testing (see paragraph 13, Operation stability, for only exception). AQL (defects per 100 units) for examination shall be 1.5 and zero for testing. Any unit found to have a defect shall be rejected.

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Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6515-01-061-7811

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACITVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

PGC

Project No. 6515-3832

Location: CID/RESUCITATOR/H12-10

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